

Font Overlap Inspection Report

Date: June 5, 2026
Inspector: Automated visual inspection
Pages Tested: Collection, Shop, Onboarding (Log Owned)
Component: WireframePeriodicTable (shared across all three pages)

Executive Summary

Significant text overflow issues were found across ALL three breakpoints. Element names (`.nm` class) overflow their cell boundaries at every responsive size. At the small breakpoint, even some element symbols (`.sym`) overflow. The root cause is that the cell widths in an 18-column grid are too narrow for the current font sizes, especially for long element names like “Praseodymium”, “Rutherfordium”, and “Darmstadtium”.

Current Font Size Configuration

Breakpoint	Screen Width	Symbol (<code>.sym</code>)	Atomic # (<code>.num</code>)	Name (<code>.nm</code>)
Small	< 768px	14px	8px	6px
Medium	768px-1023px	20px	12px	8px
Large	≥ 1024px	26px	15px	11px

Overflow Results Summary

Breakpoint	Name Overflows	Symbol Overflows	Total Elements
Small (500px)	98 / 118 (83%)	19 / 118 (16%)	118
Medium (768px)	55 / 118 (47%)	0 / 118	118
Large (1024px)	53 / 118 (45%)	0 / 118	118

Key Finding

The element name (`.nm`) text overflows in nearly half of all cells even at the largest breakpoint. At the small breakpoint, the problem is catastrophic — 83% of element names and 16% of symbols overflow.

Worst Offenders (by overflow amount)

Small Breakpoint (cell width: ~21px)

Element	Text Width	Cell Width	Overflow
Praseodymium	45px	21px	+24px
Rutherfordium	43px	21px	+22px
Darmstadtium	43px	21px	+22px
Roentgenium	40px	21px	+19px
Mendelevium	40px	21px	+19px

Medium Breakpoint (cell width: ~36px)

Element	Text Width	Cell Width	Overflow
Praseodymium	60px	36px	+24px
Rutherfordium	58px	36px	+22px
Darmstadtium	57px	36px	+21px
Mendelevium	54px	36px	+18px
Roentgenium	53px	36px	+17px

Large Breakpoint (cell width: ~50px)

Element	Text Width	Cell Width	Overflow
Praseodymium	82px	50px	+32px
Rutherfordium	79px	50px	+29px
Darmstadtium	79px	50px	+29px
Mendelevium	74px	50px	+24px
Roentgenium	73px	50px	+23px

Pages Affected

All three pages use the shared `WireframePeriodicTable` component

(`app/components/WireframePeriodicTable.jsx`) with the same CSS classes defined in `app/tailwind.css` . The overflow behavior is identical across:

1. **Collection Page** (`/app/cabinet/periodic-table`) — Names like “Rutherfordium”, “Darmstadtium”, “Meitnerium” visibly bleed outside cells
2. **Shop Page** (`/app/cabinet/shop`) — Same overflow, with shop-specific gradient styling making it slightly more noticeable
3. **Onboarding Log Owned** (`/onboarding/log-owned`) — Same component, same overflow

Visual Evidence

Collection Page (Large viewport, 1536px)

The screenshot shows the 'Luciteria Collector Cabinet' web application in a Google Chrome browser window. The address bar shows the URL `localhost:3000/app/cabinet/periodic-table`. The application interface includes a sidebar with navigation links: Dashboard, Collection (selected), Wishlist, Shop, and Notifications. The main content area is titled 'Collection' and features a status bar with the following information: Status: 1 Owned (green square), 1 Wanted (yellow square), 0 Watchlist (blue square), 116 Missing (grey square). A progress indicator shows '1 / 118 collected (1%)'. Below the status bar, a note states: 'Click an element to set its status and record acquisition notes. Elements with more than one specimen show a count badge.' The periodic table is displayed with elements from Hydrogen (1) to Oganesson (118). The bottom section contains three panels: 'Closest to Completion' showing 'Coinage Metals' at 1/4 (25%); 'Next Best Acquisitions' listing Scandium, Yttrium, and Iron; and 'Recent Activity' showing 'Au: notes updated' and 'Au: MISSING -> OWNED'.

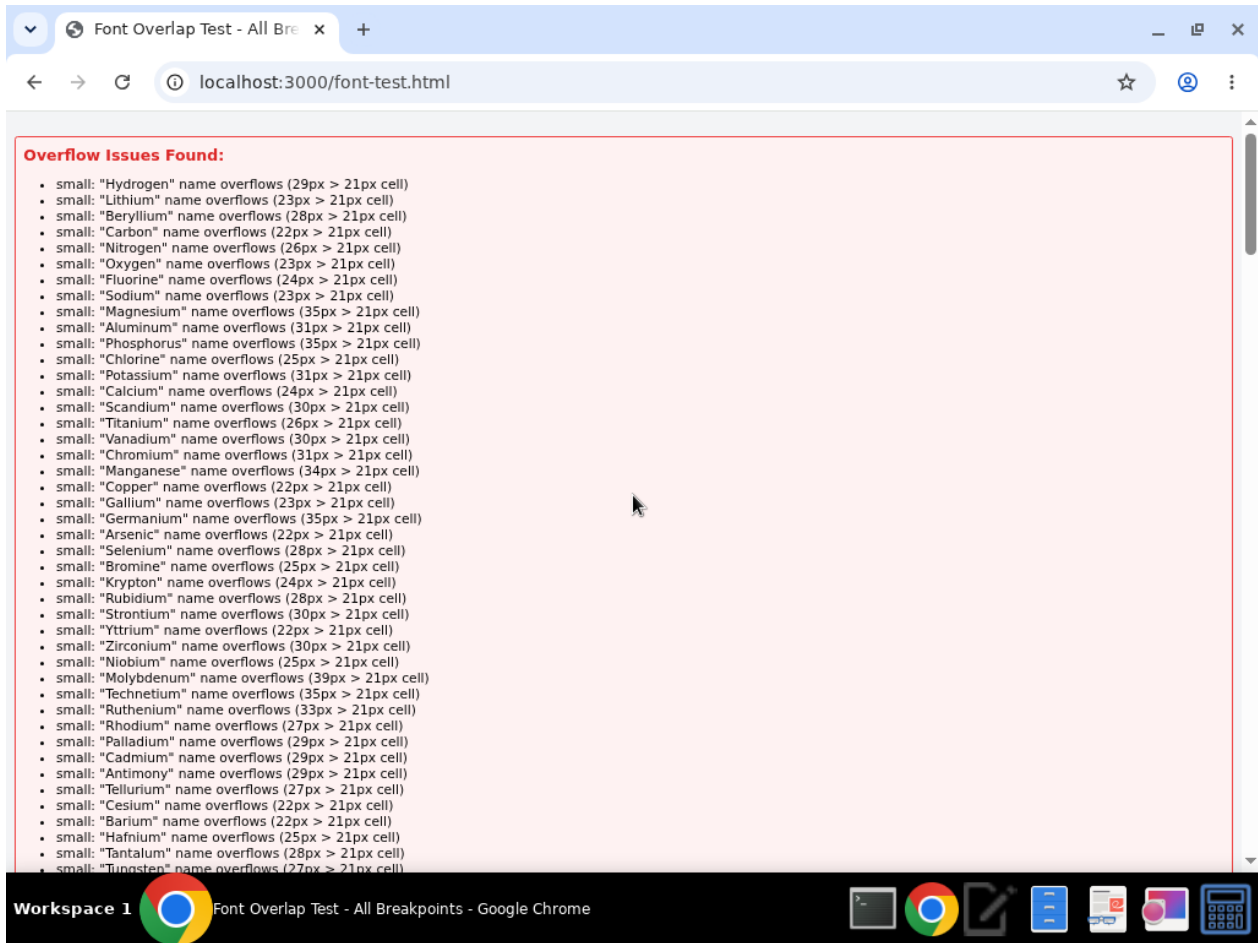
Names like “Rutherfordium”, “Molybdenum”, “Praseodymium”, “Darmstadtium” visibly overflow their cell boundaries in row 7 and the lanthanide/actinide rows.

Shop Page (Large viewport, 1536px)

The screenshot displays the 'Shop' page of the 'Luciteria Collector Cabinet' application. The browser window shows the URL 'localhost:3000/app/cabinet/shop'. The page layout includes a sidebar with navigation links: Dashboard, Collection, Wishlist, Shop (active), and Notifications. The main content area is titled 'Shop' and 'Browse Products by Format'. It features a 'Step 1: Choose Your Preferred Format' section with a 'Product Format' dropdown set to 'Lucite Cube' and a legend indicating that green elements are 'Already in your collection'. Below this is a 'Step 2: Select an element to view products' section with a prompt to 'Click any element below to open its product page on luciteria.com'. The periodic table is displayed, with elements like Gold (Au) highlighted in green. The bottom of the page shows a 'Workspace 1' bar with the Google Chrome logo and the text 'Luciteria Collector Cabinet - Google Chrome'. The taskbar at the very bottom contains various application icons.

Same overflow pattern visible, particularly in the bottom rows with longer element names.

Automated Test — All Breakpoints Side-by-Side



Red-highlighted cells indicate overflow. The medium table shows ~55 overflowing cells, and the large table shows ~53.

Medium & Large Tables with Red Highlights

Font Overlap Test - All Breakpoints - Google Chrome

localhost:3000/font-test.html

LARGE viewport (width: 1024px) — fonts: 26px/15px/11px

Periodic table of elements displayed with red highlights around the element symbols and names.

Root Cause Analysis

The fundamental issue is a **mismatch between font size and available cell width**. In an 18-column grid:

- At 500px container width: each cell is ~21px wide → 6px font names still overflow
- At 768px container width: each cell is ~36px wide → 8px font names still overflow
- At 1024px container width: each cell is ~50px wide → 11px font names still overflow

The longest element names (Praseodymium = 13 chars, Rutherfordium = 14 chars) need roughly **1.6× the cell width** even at the smallest font size.

Recommended Fixes

Option A: Add `overflow: hidden` + `text-overflow: ellipsis` (Quick Fix)

```
.cell .nm {
  overflow: hidden;
  text-overflow: ellipsis;
  white-space: nowrap;
  max-width: 100%;
}
```

This clips long names but keeps the layout clean. Most periodic tables use this approach.

Option B: Reduce Name Font Sizes Further

- Small: 6px → **4px** (may be unreadable)
- Medium: 8px → **6px**
- Large: 11px → **8px**

This reduces overflow count but won't eliminate it for the longest names.

Option C: Hide Names on Small Screens, Truncate on Medium/Large

```
/* Small: hide names entirely */
.cell .nm { display: none; }

/* Medium: show truncated */
@media (min-width: 768px) {
  .cell .nm {
    display: block;
    font-size: 6px;
    overflow: hidden;
    text-overflow: ellipsis;
    white-space: nowrap;
    max-width: 100%;
  }
}

/* Large: show with slightly larger truncated text */
@media (min-width: 1024px) {
  .cell .nm { font-size: 9px; }
}
```

Option D: Use CSS `scale()` for Names (Best Visual Result)

```
.cell .nm {
  white-space: nowrap;
  transform: scale(var(--nm-scale, 1));
  transform-origin: center;
}
```

Then use JavaScript to auto-calculate the scale factor per cell. This preserves readability while fitting within bounds.

Recommended Approach

Option A is the simplest and most standard approach for periodic tables. Combined with a `title` attribute (already present via the component), users can hover to see the full name. This is the industry-standard pattern used by most interactive periodic tables.

CSS Location

All styles are in: `app/tailwind.css` (lines 7–22)

Component: `app/components/WireframePeriodicTable.jsx`